

## **SUB-RECIPIENT AGREEMENT**

This SUB-RECIPIENT AGREEMENT (this "Agreement") is made as of January 30, 2024 by and between National Center for Defense Manufacturing and Machining having offices at 699 Scalp Avenue, Johnstown, PA 15904-1619 ("NCDMM") and Youngstown Edison Business Incubator, having offices at 241 West Federal Street, Youngstown, OH 44503 ("Sub-Recipient") with a Unique Entity ID (UEI) number of E38PN6F4AVU3 and a Cage Code of 5EAR9.

### **WITNESSETH:**

WHEREAS, NCDMM is a manufacturing technology center that has entered an agreement with the United States Department of the Air Force, dated as of November 25, 2019 under cooperative agreement number FA8650-20-2-5700 with CFDA number 12.800, to act as an administrator and technical performer in the fulfilment of the "America Makes Partnership (AMP) The National Additive Manufacturing Innovation Institute" (the "Prime Agreement");

WHEREAS, NCDMM desires to engage Sub-Recipient and America Makes (NAMII) member to assist NCDMM in satisfying certain NCDMM obligations under the Prime Agreement; and

WHEREAS, the activities contemplated by this Agreement are of mutual interest and benefit to NCDMM and Sub-Recipient, and will further the fulfilment of America Makes (NAMII);

NOW, THEREFORE, in consideration of the mutual covenants and agreements contained herein, NCDMM and Sub-Recipient hereby agree as follows:

### **ARTICLE 1. DEFINITIONS AND SCHEDULES**

1.1 **Definitions.** In addition to the terms defined elsewhere in this Agreement, the following terms shall have the meanings set forth below:

1.1.1 **"Affiliate"** means, with respect to any person or entity, any other person or entity directly or indirectly controlling, controlled by, or under common control with, such first person or entity. For purposes of this definition, the term "control" (including the correlative meanings of the terms "controlled by" and "under common control with"), as used with respect to any person or entity, shall mean the possession, directly or indirectly, of the power to direct or cause the direction of the management or policies of such person or entity, whether through the ownership of voting securities or by agreement or otherwise.

1.1.2 **"Applicable Law"** means all applicable federal, state and local laws, rules, regulations and guidelines relating to the conduct of the Parties' businesses and the performance by the Parties of their respective obligations under this Agreement.

1.1.3 **"Exploit"** means to make, have made, import, use, sell, offer for sale or otherwise dispose of, including any research, development, registration, modification, enhancement, improvement, manufacture, storage, formulation, optimization, export, transport, distribution, promotion or marketing related thereto.

1.1.5 **"Facility"** means the facility of Sub-Recipient, or any other facility used for the conduct of the activities that has been approved by NCDMM and complies with Applicable Law

1.1.6 **"Invention"** means any discovery, improvement, process, formula, invention, know-how, trade secret, procedure, device, or other intellectual property, whether or not patentable.

1.1.7 **"Materials"** means the equipment and materials supplied to Sub-Recipient by or on behalf of, or purchased at the expense of, NCDMM, reasonably necessary for Sub-Recipient to perform activities, solely as set forth in the SOW.

1.1.8 **"Party"** means either NCDMM or Sub-Recipient; **"Parties"** means both NCDMM and Sub-Recipient.

1.1.9 **"Statement of Work"** means the plan set out in Schedule A hereto.

1.1.10 **"Consortium Research"** means research supported by membership dues and / or Incorporates funds provided by the FA8650-20-2-5700 cooperative agreement. Consortium Research does not include:

1.1.10.1 Research carried out under separate contract unless specified therein, or

1.1.10.2 Research fully funded by a Member and/or without using any of the funds specified in 1.1.10 above, or

1.1.10.3 Research carried out independently by a Member using America Makes (NAMII) facilities

1.1.11 **"Consortium Developed Intellectual Property"** means individually and collectively all Intellectual Property which are conceived or made solely or jointly by Member(s) during the performance of this sub award project under Consortium Research.

1.1.12 **"Background Intellectual Property"** means technical know-how, inventions, technical data, discoveries, materials, samples, software, software programs, whether patentable or not, copyrightable programs, documentation and reports whether in existence at the time of this agreement or coming into existence subsequent to this agreement, which were not developed in the course of performance of any sub award under the America Makes (NAMII) consortium.

1.1.13 **"Intellectual Property"** means technical know-how, inventions, technical data, discoveries, materials samples, software, software programs, documentation, reports, any and all other copyrightable materials, and/or invention or discovery that is or may be patentable or otherwise protectable under title 35 of the U.S. Code.

1.2 Schedules. The Schedules attached hereto are incorporated in and are deemed to be an integral part of this Agreement.

## **ARTICLE 2. ACTIVITIES**

2.1 . Sub-Recipient shall undertake and perform work in accordance with the Statement of Work (SOW) attached hereto as Schedule A, as the same may be amended from time to time in accordance with the terms hereof (Sub-Recipient's responsibilities as set out in the SOW being hereinafter referred to as the "Activities"). The specific objectives, scope, deliverables and estimated schedule for the Activities are set forth in the SOW. Sub-Recipient agrees that it shall implement changes to the SOW that are reasonably requested in writing by NCDMM, subject, if applicable, to NCDMM's payment of any reasonable additional fees and/or expenses incurred by Sub-Recipient as determined in accordance with Section 4.4 that arise solely from such changes to the SOW. Any material change in the SOW shall be evidenced in a written amendment of Schedule A.

2.2 Cooperation. The Parties shall cooperate with each other in the performance of this Agreement and shall deal honestly and in good faith with each other.

2.3 Reporting/Transfer of Results. Sub-Recipient shall comply with the monthly written progress report requirements set forth in Section 4.5.1 and, if requested by NCDMM, on an ongoing basis, Sub-Recipient shall keep NCDMM informed of progress and interim results on an informal basis, including if requested, periodic meetings to discuss Activities progress and interim results. Further specific guidance regarding programmatic requirements are found in Schedule D. Upon completion of the Activities, termination of this Agreement pursuant to Section 10.2, or expiration of this Agreement (without a further agreement related to the substance hereof entered into as contemplated in Section 10.1), Sub-Recipient will provide NCDMM with a comprehensive written report of the results that have been developed, compiled or learned during the course of the Activities, and will comply with NCDMM's reasonable requests for follow-up information.

### **ARTICLE 3. SUPPLY OF MATERIALS AND INFORMATION**

3.1 Supply or Purchase of Materials. If applicable, subject to the final sentence of this Section 3.1, within a commercially practicable time after written request from Sub-Recipient, NCDMM shall supply Sub-Recipient, if previously negotiated, Materials as set forth in the SOW. Sub-Recipient shall own all right, title and interest in and to any Materials, unless specifically identified to the contrary in writing. If NCDMM supplies Materials to Sub-Recipient, NCDMM shall provide Sub-Recipient with written documentation evidencing the value of such Materials (the "Valuation Price"). If the Parties disagree over the Valuation Price provided by NCDMM, after thirty (30) days of good faith discussions, either Party may request that the matter be submitted to arbitration for a binding decision in accordance with Section 4.7.

3.2 Information. If requested by Sub-Recipient to provide information in NCDMM's possession that is not otherwise accessible to Sub-Recipient and that is reasonable and necessary for Sub-Recipient's performance under this Agreement ("Information"), subject to the following sentence, NCDMM shall provide such Information (or an explanation of the legitimate reason for any refusal or delay, and if delay, a projected date by which such Information will be provided) within a commercially practicable period of time after receipt of Sub-Recipient's written request. Notwithstanding the foregoing, NCDMM may, in its sole discretion, refuse to provide any Information to Sub-Recipient, including Confidential Information, Inventions or any other Information NCDMM deems, in its sole discretion, to be proprietary or unnecessary for use in the Activities, provided that Sub-Recipient shall not be deemed in breach of this Agreement if the sole reason it cannot perform any Research or Development Activities is NCDMM's refusal to provide Information reasonable and necessary for the conduct of the Activities. NCDMM shall own all right, title and interest in and to any Information, unless specified to the contrary in writing.

3.3 Use of Materials and Information. Sub-Recipient shall (i) use the Materials and Information only for the purposes described in the SOW or such other purposes as NCDMM may approve in writing, (ii) restrict access to and use of the Materials and Information to the Principal Investigator and other researchers employed by, or who are agents of, Sub-Recipient or any permitted Sub-Recipients for whom such access and use is required to conduct the Activities

and who are bound by obligations of confidentiality at least as strict as those set forth in Section 6.

3.4 Supply of Other Materials by Sub-Recipient. All equipment and materials other than the Materials set forth in the SOW, required for the Activities shall be purchased and paid for by Sub-Recipient, unless otherwise agreed to in writing by NCDMM. Sub-Recipient shall own all right, title and interest in and to such equipment and materials purchased under this Section 3.4, unless specifically identified to the contrary in writing.

3.5 Non-Exclusive License. If, in its sole discretion, NCDMM determines that it is necessary for Sub-Recipient's conduct of the Activities, NCDMM may, by separate written agreement with Sub-Recipient, grant Sub-Recipient a non-exclusive, non-transferable and non-assignable right and license to use an Invention owned by, or licensed (with the right to sublicense) to, NCDMM, which license shall be limited in duration to the Term or such shorter period of time designated by NCDMM, limited in territory to use in a Facility, and limited in scope to use solely as needed to perform the Activities (or a part thereof as designated by NCDMM). For the avoidance of doubt, this Agreement does not grant any such license.

#### **ARTICLE 4. BUDGET AND PAYMENT**

4.1 Compensation and Payment Terms. NCDMM is entering into a Cost Reimbursement No Fee Agreement with the Sub-Recipient in accordance with the approval of AFRL found in Schedule B. In consideration for the satisfactory performance of the Activities by Sub-Recipient in accordance with this Agreement, subject to the provisions of this Section 4, NCDMM shall, within forty-five (45) to sixty (60) days of receipt of an invoice and any supporting documentation in accordance with Section 4.5, pay to Sub-Recipient the amounts for the invoiced incurred expenses. The Parties agree that NCDMM shall pay no Amounts or Expenses until it has been funded under the Prime Contract with respect to the Research and Development Activities. All payments hereunder shall be made in U.S. dollars, by check.

4.2 Reimbursement of Expenses. NCDMM shall reimburse Sub-Recipient for actual expenses incurred by Sub-Recipient for reasonable, necessary and verifiable:

4.2.1 Materials that are set forth in the SOW that NCDMM does not provide to Sub-Recipient;

4.2.2 Travel expenses incurred while performing the Activities in accordance with the Joint Travel Regulations as published by the Federal Government("Travel Expenses");

4.2.3 Other expenses expressly set forth in Schedule B (Budget) ("Other Approved Expenses");

4.2.4 Any other expenses specifically approved in advance in writing by NCDMM.

Collectively, the expenses referred to in Sections 4.2.1 through 4.2.4, are referred to herein as the "Expenses." Notwithstanding the foregoing, Sub-Recipient shall only be reimbursed for Expenses that are allowable costs under 2 Code of Federal Regulations 200, Subpart E. Sub-Recipient acknowledges and agrees that NCDMM shall not pay or reimburse Sub-Recipient for any Expenses incurred by, or purchases made by or on behalf of, Sub-Recipient other than those set forth in this Section 4.2.

4.3 Total Obligation. Unless otherwise agreed in writing by the Parties, the foregoing amounts represent NCDMM's total obligation for any and all Activities and any and all expenses incurred by or on behalf of Sub-Recipient. The total funds authorized by this agreement shall not exceed \$1,103,594.

4.4 Expense Changes. The Expenses are based on the scope of work and assumptions stated in the SOW. If NCDMM requests a change in the SOW which the Parties agree, in good faith, would result in an increase to Sub-Recipient of its cost of conducting the Activities, the Parties shall, in good faith, evaluate and discuss revised Expenses. Based on this evaluation and discussion, the Parties shall amend the SOW and Budget by written agreement. In the event the Parties cannot agree on a price change after thirty (30) days of good faith discussions, either Party may request that the matter be submitted to arbitration for a binding decision in accordance with Section 4.7.

#### 4.5 Invoicing and Progress Reports.

4.5.1 Sub-Recipient shall invoice NCDMM once each month during the Term. Each invoice shall be accompanied by a written progress report, which shall describe the Activities Sub-Recipient has performed since the date of the last invoice and provide such other information as may be required by this Agreement or reasonably requested by NCDMM. Each monthly invoice shall include expenses and cost share for expenses incurred to NCDMM's reasonable satisfaction due Sub-Recipient since the date of the last invoice. For Expenses to be reimbursed pursuant to Section 4.2, Sub-Recipient shall include in the monthly invoice a summary report of the Expenses incurred since the date of the last invoice.

4.5.2 If NCDMM disputes any portion of an invoice, it shall pay the undisputed portion and shall provide Sub-Recipient with written notice of the disputed portion and its reasons therefor, and NCDMM shall not be obligated to pay interest on such disputed portion. In the event the Parties cannot agree on an invoice after thirty (30) days of good faith discussions, either Party may request that the matter be submitted to arbitration for a binding decision in accordance with Section 4.7.

4.5.3 **Closeout, Adjustment, Continuing Responsibilities and Reimbursement** – Final payment will not be made nor can this agreement be closed out until the sub-recipient delivers to NCDMM all the final closeout paperwork, disclosure of subject inventions, and the final project report. NCDMM may adjust downward the total obligation for the effort if this information is not provided.

#### 4.6 Records Retention and Audit.

4.6.1 Records Retention. Sub-Recipient shall keep or cause to be kept accurate records or books of account in accordance with applicable generally accepted accounting principles that, in reasonable detail, fairly reflect the invoiced Fees and Expenses. Such books and records shall be maintained by Sub-Recipient for at least three (3) years following the end of the calendar year to which they pertain.

4.6.2 Audit. Upon the written request of the United States Government or NCDMM, Sub-Recipient shall permit a Government auditor, or a certified public accountant or a person possessing similar professional status and associated with an independent accounting firm reasonably acceptable to the Parties, to inspect during regular business hours all or any part of Sub-Recipient's records and books necessary to verify invoices related to Expenses and Cost Share.

4.7 Arbitration. In the event that a Party requests a Valuation Price determination in accordance with Section 3.1, a price change determination in accordance with Section 4.4, or an invoice determination in accordance with Section 4.5.2 to be submitted to an arbitrator, such determination shall be determined by a single arbitrator in accordance with the Commercial Arbitration Rules of the American Arbitration Association. The decision of the arbitrator shall be final and conclusive, absent manifest error. The Parties shall amend this Agreement if and as appropriate to implement the decision of the arbitrator.

## **ARTICLE 5. REPRESENTATIONS, WARRANTIES AND COVENANTS**

5.1 Sub-Recipient represents and warrants to, and covenants with, NCDMM that:

5.1.1 Sub-Recipient shall perform the Activities in good scientific manner and in compliance with Applicable Law and shall use reasonable efforts to perform the Activities in accordance with the SOW.

5.1.2 Neither Sub-Recipient nor any of its principal officers or employees has been debarred or suspended by any governmental entity from conducting any Activities or is subject to any such debarment and Sub-Recipient will not use in any capacity, in connection with its obligations under this Agreement, including via any permitted Sub-Recipient, any person who has been so debarred.

5.1.3 Each Facility is and shall be during the Term in material compliance with all requirements of Applicable Law. There are, and during the Term shall be, no pending or uncorrected citations or adverse conditions noted in any governmental inspection of any Facility with respect to any Activities.

5.1.4 Sub-Recipient owns or has the legal right to use all patents, know-how and other intellectual property rights required for use in its conduct of the Activities, excluding those rights obtained from NCDMM under separate written agreement of the Parties or owned by or licensed to NCDMM pursuant to this Agreement or otherwise.

5.1.5 The Principal Investigator is, and at all times during the conduct of the Activities shall be, qualified by training and experience with appropriate expertise to conduct and supervise the Activities.

5.1.6 Sub-Recipient (for itself and each Facility) and the Principal Investigator have, and at all times during the conduct of the Activities shall have, all appropriate licenses, approvals, authorizations, registrations, permits, and certifications necessary to safely, adequately and lawfully perform the Activities. Sub-Recipient shall provide copies of all such licenses, approvals, authorizations, registrations, permits or certifications upon NCDMM's request.

Sub-Recipient shall advise NCDMM immediately if any of the foregoing in this Section 5.1 ceases to be true and correct.

5.2 Each Party represents and warrants to, and covenants with, the other Party that:

5.2.1 It has the requisite power and authority to execute, deliver and perform its obligations under this Agreement and all governmental and third-party approvals necessary for it to enter into and perform its obligations under this Agreement have been obtained, are in full force and effect and are final and non-appealable.

5.2.2 This Agreement is its legal, valid, and binding obligation, enforceable in accordance with its terms and conditions.

5.2.3 The execution and delivery of this Agreement and the transactions contemplated in it do not violate, conflict with, or constitute a default under the terms or provisions of any agreement or other instrument to which it is a party or by which it is bound, or any order, award, judgment, or decree to which it is a party or by which it is bound.

5.2.4 It is in compliance with all Applicable Law and it will comply with all Applicable Law during the Term.

A Party shall immediately notify the other Party if any of the foregoing in this Section 5.2 with respect to such first Party ceases to be true or correct.

5.3 Disclaimer. EXCEPT AS EXPRESSLY SET FORTH IN THIS SECTION 5, NCDMM MAKES NO REPRESENTATIONS AND GRANTS NO WARRANTIES, EXPRESS OR IMPLIED, EITHER IN FACT OR BY OPERATION OF LAW, BY STATUTE OR OTHERWISE, UNDER THIS AGREEMENT, AND NCDMM SPECIFICALLY DISCLAIMS ANY OTHER WARRANTIES, WHETHER WRITTEN OR ORAL, OR EXPRESS OR IMPLIED, INCLUDING ANY WARRANTY OF QUALITY, MERCHANTABILITY OR FITNESS FOR A PARTICULAR USE OR PURPOSE WITH RESPECT TO ANY MATERIALS (AS DEFINED IN SECTION 1.1.7) OR ANY WARRANTY AS TO THE NON-INFRINGEMENT OF ANY INTELLECTUAL PROPERTY RIGHTS OF THIRD PARTIES UNDER THIS AGREEMENT.

## **ARTICLE 6. CONFIDENTIALITY**

6.1 In the normal and routine completion of this project, there may be some occasions where one Party discloses information that is proprietary and confidential to the disclosing Party. All such information which a disclosing Party believes to be proprietary, confidential, trade secret or the like undisclosed information shall be provided in writing or other permanent visual form (such as, for example, a physical prototype, physical sample, photograph, video or sound recording on any type of recording media) and designated as proprietary with an appropriate restrictive legend stamped or marked thereon at the time disclosure. Proprietary Information disclosed orally, must be identified as proprietary at the time of disclosure and summarized in writing as being proprietary within thirty (30) days of such disclosure.

6.2 The receiving Party agrees to exercise the same care and safeguards with respect to Proprietary Information disclosed by the disclosing Party as used to maintain the confidentiality of its own information of like character, but in no event less than a reasonable degree of care.

6.3 The obligations of confidentiality and non-disclosure set forth in this section 6 shall continue for a period of five (5) years from the termination or expiration of this Agreement or until the information ceases to fall under the definition of Proprietary Information.

## **ARTICLE 7. INTELLECTUAL PROPERTY**

7.1 **Background Intellectual Property.** Each Sub-Recipient/Sub-Recipient/Member shall retain all rights to its Background Intellectual Property; and the decision to make available any such Background Intellectual Property for use in a Sub-Recipient/Member's sub award project shall be at the sole discretion of each Sub-Recipient/Member and in accordance with DOD regulations with respect to identification of all such Background Intellectual Property. No license or rights are granted to a Sub-Recipient/Member's Background Intellectual Property under this Agreement.

7.1.1 In the event that one Sub-Recipient/Member may require use of another Sub-Recipient/Member's Background Intellectual Property that has been disclosed by a Sub-Recipient/Member as part of the Consortium Research in order to successfully commercialize any CDIP then the Sub-Recipient/Members agree to discuss potential licensing terms and

conditions in a separate legally binding agreement between the Sub-Recipient/Members, separate from this Agreement. Sub-Recipient/Member(s) are not required to license any such originating Sub-Recipient/Member(s) Background Intellectual Property.

**7.2 Consortium Developed IP** shall be owned by the respective inventing or creating organizations, subject to any government rights and/or any pre-existing rights of any third party and subject to the following conditions:

7.2.1 If a Party solely or jointly creates CDIP, the Party must disclose the creation of such CDIP to its technology transfer office, licensing office or other similar department ("Party's TechTransfer Office"). A non-confidential summary of the CDIP disclosed to the Party's TechTransfer Office shall be sent to NCDMM as soon as practicable so that NCDMM can maintain a list of CDIP. Sub-Recipient/Members owning CDIP shall grant upon request to NCDMM and Sub-Recipient/Members-In-Good-Standing at the time of creation a limited, non-exclusive, royalty-free license to use the CDIP for the Sub-Recipient/Member's internal procedures, research or development purposes (but not to make, use, or sell products or external processes for commercial purposes, with the exception of licenses granted pursuant to Section 7.2.3). Such licenses shall be granted to interested Sub-Recipient/Members upon request in a separate legally binding mutually agreeable license agreement between the Sub-Recipient/Members. Payment of patent expenses may be required of Parties granted non-exclusive, royalty-free commercial licenses by universities and other non-profit institutions. Such licenses for Sub-Recipient/Members shall be without the right to grant sublicenses to third parties, except for any Sub-Recipient/Member-designated agents, contractors and non-employee students ("Permitted Third Parties") performing work for the benefit of such Sub-Recipient/Member. Under these circumstances the Sub-Recipient/Member is responsible for having any and all appropriate written agreements with such Permitted Third Parties to enable Sub-Recipient/Member's compliance with this Agreement and is responsible for such parties' use of the CDIP in the same manner Sub-Recipient/Member is responsible for its own use of such CDIP (e.g., violation of the license parameters set forth in this section by a Sub-Recipient/Member's Permitted Third Parties shall be considered a breach of this Agreement by Sub-Recipient/Member).

7.2.2 Sub-Recipient/Members are strongly encouraged to seek legal protection in the form of patents as soon as is reasonably possible following disclosure of all CDIP to NCDMM. Protection of a Sub-Recipient/Member's solely developed CDIP shall be done at Sub-Recipient/Member's own expense and through use of their respective Party's TechTransfer Office. Each Sub-Recipient/Member agrees to notify in writing NCDMM in a timely manner of all such actions in which legal protection is or has been sought so that NCDMM can enter such information in its invention disclosure database. With respect to jointly developed CDIP the relevant Sub-Recipient/Members agree to negotiate a separate legally binding agreement encompassing those terms and conditions to be used to govern the manner in which jointly developed CDIP will be owned, administered, protected, and licensed. NCDMM will be notified in writing in a timely manner of the existence of these agreements between Sub-Recipient/Members and NCDMM shall maintain pertinent information in its invention disclosure database. In the event that a single Sub-Recipient/Member for solely developed CDIP, or all Sub-Recipient/Members with an ownership right for jointly developed CDIP choose not to seek legal protection and thereby elect not to file a patent application on any

CDIP, then Sub-Recipient/Member(s) agree to notify NCDMM in writing of its intent and must report any pending publication or presentation to NCDMM at the time of this notification. NCDMM may negotiate to obtain such protection at its own expense where Sub-Recipient/Member(s) choose not to seek legal protection. Ownership of CDIP shall remain with the originating Sub-Recipient/Member(s).

7.2.3 It is anticipated that one of the outcomes of an active IP licensing and commercialization plan is the generation of royalty income by a respective Sub-Recipient/Member. It is acknowledged that Sub-Recipient/Members of academic, government, and industry sectors will manage the disposition and reporting requirements of all royalties received in accordance with their institution's existing policies, through their Party's TechTransfer Office. To the extent it may legally do so, Sub-Recipient/Members owning CDIP shall grant NCDMM and Platinum Sub-Recipient/Members-In-Good-Standing a limited, non-exclusive, royalty-free license to use the CDIP for commercial purposes and Gold Sub-Recipient/Members-In-Good-Standing shall be granted such licenses at a fair market value royalty rate. Such licenses shall be granted to interested Sub-Recipient/Members upon request in a separate legally binding mutually agreeable agreement between the Sub-Recipient/Members. Such licenses for Sub-Recipient/Members shall be without the right to grant sublicenses to third parties, except for any Sub-Recipient/Member-designated agents, contractors and non-employee students performing work for the benefit of such Sub-Recipient/Member, provided, however, the Sub-Recipient/Member is responsible for having any and all appropriate written agreements with such parties to enable Sub-Recipient/Member's compliance with this Agreement and is responsible for such parties' use of the CDIP in the same manner Sub-Recipient/Member is responsible for its own use of such CDIP (e.g., violation of the license parameters set forth in this section by a Sub-Recipient/Member's contractor shall be considered a breach of this Agreement by Sub-Recipient/Member). In addition, sublicensing shall be permitted to the licensee's direct customers.

7.3 Licenses granted under this Section 7 shall be subject to these additional terms:

Any licenses granted to a Sub-Recipient/Member through a separate document as specified in this Section 7 shall become a royalty bearing license at fair market value for such a Sub-Recipient/Member who is designated a Withdrawn Sub-Recipient/Member.

It is understood that the United States Government (through any of its agencies or otherwise) may provide use of its facilities or equipment and / or may provide funds for Consortium Research. As a result, this Agreement, any and all rights and obligations of the Sub-Recipient/Members to any CDIP resulting from use of any United States Government's facilities, equipment or funds are subject to any and all applicable rights of the United States Government.

Notwithstanding anything to the contrary in this Agreement, certain laws, regulations and/or policies may prevent and/or limit certain Sub-Recipient/Members' ability to offer royalty-bearing licenses to CDIP that has previously been licensed by such Sub-Recipient/Members on a royalty-free basis. Therefore, the ability to charge royalties to Gold Sub-Recipient/Members, Silver Sub-Recipient/Members, and/or third parties is subject to the granting Sub-Recipient/Member's ability to do so in light of then-existing contractual obligations, legal and regulatory requirements, and policies of the granting Sub-Recipient/Member.

## **ARTICLE 8. INDEMNIFICATION AND INSURANCE**

8.1 **Indemnification by Sub-Recipient.** Sub-Recipient agrees to defend, indemnify and hold harmless NCDMM and its Affiliates and their respective directors, officers, employees and agents from and against any and all costs (including reasonable legal fees), damages, expenses and losses (collectively, "Losses") arising from any third party suit, claim, demand, assessment, action or proceeding (collectively, "Claims"), in any manner caused by, resulting from or arising out of any misrepresentation, negligence or breach of this Agreement on the part of Sub-Recipient, any of Sub-Recipient's Affiliates or any permitted Sub-Recipients, or any of Sub-Recipient's or its Affiliates' or permitted Sub-Recipients' directors, officers, employees or agents, including Losses arising out of the breach or inaccuracy of any of Sub-Recipient's representations, warranties, or covenants under Section 5 of this Agreement, except to the extent any such Loss is attributable to any breach by NCDMM of this Agreement, or any negligence, wilful misconduct or unreasonable inaction by NCDMM, any of NCDMM's Affiliates, or any or NCDMM's or its Affiliates' directors, officers, employees or agents.

8.2 **Indemnification by NCDMM.** NCDMM agrees to defend, indemnify and hold harmless Sub-Recipient and its Affiliates and their respective directors, officers, employees and agents from and against any and all Losses arising from any third party Claims, in any manner caused by, resulting from or arising out of any misrepresentation, negligence or breach of this Agreement on the part of NCDMM, or any of NCDMM's directors, officers, employees or agents, including any Losses arising out of any breach or misrepresentation made by NCDMM under Section 6, except to the extent any such Loss is attributable to any breach by Sub-Recipient of this Agreement, or any negligence, wilful misconduct or inaction by Sub-Recipient, any of Sub-Recipient's Affiliates or any permitted Sub-Recipients, or any of Sub-Recipient's or its Affiliates' or any permitted Sub-Recipients' directors, officers, employees or agents.

8.3 **Indemnification Procedures.** In the event that either Party (an "Indemnified Party") seeks indemnification from the other Party (the "Indemnifying Party") under the terms of this Article 8 with respect to any claim made by a third party, it shall inform the Indemnifying Party of the claim as soon as reasonably practicable after it receives notice thereof, shall permit the Indemnifying Party, at the Indemnifying Party's cost, to assume direction and control of the defense of the claim, and shall co-operate as requested (at the expense of the Indemnifying Party), in the defense of the claim. After notice to the Indemnified Party of the Indemnifying Party's election to assume the defense of such claim, the Indemnifying Party shall be liable to the Indemnified Party only for such legal or other expenses subsequently incurred by the Indemnified Party in connection with the defense thereof at the request of the Indemnifying Party. The Indemnifying Party shall not settle or otherwise compromise any claim or suit without the prior written consent of the Indemnified Party, which shall not be unreasonably withheld. As to those third party claims with respect to the which the Indemnifying Party does not elect to assume control of the defense, the Indemnified Party will afford the Indemnifying Party an opportunity to participate in such defense, at the Indemnifying Party's own cost and expense, and will not consent to the entry of any judgment, settle or otherwise dispose of any of the same without the consent of the Indemnifying Party, which shall not be unreasonably withheld.

8.4 Limitation of Liability and Claims. NCDMM shall not be liable to Sub-Recipient hereunder for any indirect, incidental, special, punitive or consequential damages of any kind, including lost profits or loss of goodwill or otherwise; and in no event shall the collective, aggregate liability of NCDMM and its Affiliates and its and their respective directors, officers, employees and agents under this Agreement exceed the amount of compensation actually received by Sub-Recipient from NCDMM pursuant to this Agreement. No action, regardless of form, arising out of or in any way connected with this Agreement or the Activities, may be brought by Sub-Recipient more than one (1) year after the cause of action accrued.

## **ARTICLE 9. PUBLICATIONS**

9.1 The Parties agree and expect that results of Consortium Research associated with this agreement shall be published or otherwise made publicly available and that Parties engaged in Consortium Research shall be permitted to present at symposia, national or regional professional meetings and to publish in journals, theses or dissertations, or by other means of their own choosing, the results of their research, provided that nothing will be done which could bar the availability of patent protection with respect to CDIP of a Sub-Recipient/Member or America Makes (NAMII) invention or which would disclose Proprietary Information of any Sub-Recipient/Member or of America Makes (NAMII) or disclose information in violation of the applicable U.S. laws and regulations (e.g., the International Traffic in Arms Regulations ("ITAR") and the Export Administration Regulations ("EAR") that govern the export of specific technical data and technologies, including software, prototypes and other intellectual property, to foreign countries and foreign nationals ("Export Control Laws").

9.1.1 A Party will not make a public disclosure without a review of the full text of the proposed publication, presentation or other form of public disclosure by the Sub-Recipient/Member(s) involved, the America Makes (NAMII) Director and Government PM as described below. The Sub-Recipient/Member(s) involved, the America Makes (NAMII) Director and Government PM shall be provided a copy of the proposed public disclosure at least sixty-five (65) days in advance of the submission of such proposed public disclosure and shall have two (2) weeks after receipt of said proposed disclosure to respond in writing to the submitting Party to identify Proprietary Information and/or to identify any potentially patentable CDIP and/or to identify any CDIP in which the submitting Party does not have an ownership interest. A submitting Party agrees to remove any identified Proprietary Information, potentially patentable CDIP and/or CDIP in which the submitting Party does not have an ownership prior to public disclosure (or, for potentially patentable CDIP in which the submitting Party does have an ownership interest, delay public disclosure for a period of sixty (60) days from the date of the response).

9.1.2 Notwithstanding anything to the contrary above, student theses and dissertations shall be subject to a separate review and comment process wherein the student shall submit such student thesis or dissertation in draft form at least sixty (60) days in advance of the date of their final defense in order to afford an opportunity to identify Proprietary Information and/or identify any potentially patentable CDIP and/or any CDIP in which the Party's student does not have an ownership interest.

9.1.3 America Makes (NAMII) and/or America Makes (NAMII) Sub-Recipient/Members may negotiate and implement a more restrictive public disclosure agreement than defined in paragraphs 9.1.1 and 9.1.2 for a specific America Makes (NAMII) project based on the need for extended non-disclosure of CDIP by the project participants.

9.1.4 An acknowledgment of funding and a disclaimer shall appear in the publication of any material, whether copyrighted or not, resulting from an America Makes (NAMII) project incorporating U.S. Government funds granted in support of the America Makes (NAMII) Consortium. The acknowledgement shall read:

“This material is based on research sponsored by Air Force Research Laboratory under agreement number FA8650-20-2-5700. The U.S. Government is authorized to reproduce and distribute reprints for Governmental purposes notwithstanding any copyright notation thereon.”

The disclaimer shall read:

“The views and conclusions contained herein are those of the authors and should not be interpreted as necessarily representing the official policies or endorsements, either expressed or implied, of Air Force Research Laboratory or the U.S. Government.”

## **ARTICLE 10. TERM AND TERMINATION**

10.1 Term. The term of this Agreement shall begin on the date hereof and shall continue until January 4, 2026 (the “Term”). Subject to termination in accordance with Section 10.2, the Parties agree to discuss in good faith prior to the end of the Term amending this Agreement (including this Section 10.1 and Schedules A, B, C, and D (if applicable)) or entering into a separate agreement governing the Parties’ relationship with respect to the general substance of this Agreement, subject to mutual agreement, for one or more years following the end of the Term.

### **10.2 Termination.**

10.2.1 Either Party shall have the right to terminate this Agreement, by giving written notice to the other Party, upon material breach by the other Party of any material provision of this Agreement, including any covenant, representation or warranty contained in this Agreement or failure to perform any material portion of the Activities in accordance with the SOW, provided that such breach shall have continued for a period of thirty (30) days after the non-breaching Party has delivered written notice of said breach to the breaching Party.

10.2.2 This Agreement may be terminated at any time by written agreement of the Parties.

10.2.3 NCDMM shall have the right to terminate this Agreement for any or no reason upon thirty (30) days prior written notice to Sub-Recipient.

10.2.4 NCDMM shall have the right to terminate this Agreement immediately upon termination of the Prime Agreement (in whole or part) or upon any amendment or other change to the Prime Agreement that materially affects the need for or value of (each in NCDMM’s reasonable sole discretion) all or a portion of the Activities hereunder.

10.2.5 Sub-Recipient shall have the right to terminate this Agreement upon thirty (30) days written notice if circumstances beyond its reasonable control preclude continuation of the Activities as described in the SOW.

10.3 Rights and Duties Upon Termination or Expiration. Upon termination or expiration (without a further agreement related to the substance hereof entered into as contemplated in

Section 10.1) of this Agreement, (i) Sub-Recipient shall cease all work on the Activities and turn over to NCDMM all documentation prepared by (or on behalf of, as the case may be) Sub-Recipient for NCDMM during the Activities (whether in written or electronic form) and (ii) NCDMM shall, in accordance with Article 4 pay Sub-Recipient. Termination or expiration of this Agreement shall be without prejudice to any rights that shall have accrued to the benefit of a Party prior to such termination or expiration.

10.4 Survival. The respective rights and obligations set forth in the previous Articles and any other provisions required to interpret and enforce the Parties' rights and obligations under this Agreement shall indefinitely survive the expiration or termination of this Agreement to the extent expressly set forth therein or if no survival term is expressly set forth therein to the extent necessary to preserve these rights and obligations.

#### **ARTICLE 11. MISCELLANEOUS**

11.1 No Commercialization. Sub-Recipient agrees that it shall not commercialize or, except as expressly permitted under the terms of this Agreement, otherwise Exploit any Materials, Information or Inventions owned by or licensed to NCDMM.

11.2 Equitable Relief. Sub-Recipient acknowledges and agrees that the restrictions set forth in Articles 6 and 7 and Sections 8.1 and 8.2 of this Agreement are reasonable and necessary to protect the legitimate interests of NCDMM and that NCDMM would not have entered into this Agreement in the absence of such restrictions, and that any violation or threatened violation of any provision of Article 6 or 7 or Section 8.1 or 8.2 will result in irreparable injury to NCDMM. Sub-Recipient also acknowledges and agrees that in the event of a violation or threatened violation of any provision of Article 6 and 7 or Section 8.1 or 8.2, NCDMM shall be entitled to preliminary and permanent injunctive relief, without the necessity of proving irreparable injury or actual damages and without the necessity of having to post a bond. The rights provided in the immediately preceding sentence shall be cumulative and in addition to any other rights or remedies that may be available to NCDMM. Nothing in this Section 11.2 is intended, or should be construed, to limit NCDMM's right to preliminary and permanent injunctive relief or any other remedy for a breach of any other provision of this Agreement.

11.3 Publication. Sub-Recipient, including its Affiliates and any permitted Sub-Recipients and its or their respective employees or agents, has no right to publish the NCDMM name or logo, except as expressly consented to in writing by NCDMM.

11.4 Waiver. The failure of a Party to enforce any breach or provision of this Agreement shall not constitute a continuing waiver of such breach or provision and, subject to Section 8.4, such Party may at any time thereafter act upon or enforce such breach or provision of this Agreement. Any waiver of breach executed by either Party shall affect only the specific breach and shall not operate as a waiver of any subsequent or preceding breach. To be effective any waiver must be in writing.

11.5 Assignment and Foreign Participation. Sub-Recipient shall not sell, transfer, assign, pledge or otherwise dispose of, whether voluntarily, involuntarily, by operation of law or otherwise, this Agreement or any of its rights or obligations hereunder without the prior written consent of NCDMM, provided that in the event such consent and approvals, if applicable, are granted, the proposed assignee shall agree in writing, in a form reasonably satisfactory to

NCDMM, to perform all of Sub-Recipient's obligations hereunder. Any attempt to assign, transfer, subcontract or delegate any portion of this Agreement or Sub-Recipient's rights or obligations hereunder in violation of this Section 11.5 shall be null and void. Subject to the terms hereof, this Agreement shall be binding upon and inure to the benefit of the successors and permitted assigns of the respective Parties hereto.

Any foreign person or other foreign participation conducting work under this effort in or outside of the United States must be approved prior to any of the work being completed. Requests for approval should be submitted to the cognizant NCDMM project manager managing the effort. The request will be submitted to the Government program office for final review and approval. No work can be completed outside of the United States without this approval.

11.6 Severability. If any provision of this Agreement is held to be illegal, invalid or unenforceable under any present or future law, and if the rights or obligations of any Party under this Agreement will not be materially and adversely affected thereby, (a) such provision shall be fully severable, (b) this Agreement shall be construed and enforced as if such illegal, invalid or unenforceable provision had never comprised a part hereof, (c) the remaining provisions of this Agreement shall remain in full force and effect and shall not be affected by the illegal, invalid or unenforceable provision or by its severance herefrom and (d) in lieu of such illegal, invalid or unenforceable provision, there shall be added automatically as a part of this Agreement, a legal, valid and enforceable provision as similar in terms to such illegal, invalid or unenforceable provision as may be possible and reasonably acceptable to the Parties.

11.7 Notices. Any notice, request or other communication required to be given pursuant to the provisions of this Agreement shall be in writing and shall be deemed to be given when delivered in person, by facsimile or by courier (return receipt requested) or five (5) days after being deposited in the United States mail, postage prepaid, certified, return receipt requested to the Parties addressed as follows:

| If to NCDMM, to:                 | If to Sub-Recipient, to: |
|----------------------------------|--------------------------|
| Gary R. Fleegle, President & CEO | Barbara Ewing, Contracts |
| 699 Scalp Avenue                 | 241 W Federal Street     |
| Johnstown, PA 15904-1619         | Youngstown, OH 44503     |
| Tel: (724) 539-1361              | Tel: (330) 746-5003      |
| gary.fleegle@ncdmm.org           | bewing@ybi.org           |

Either Party may change its contact person, address or telephone or facsimile number by giving the other Party written notice, delivered in accordance with this Section 10.7.

11.8 Construction. Except where the context otherwise requires, wherever used, the singular shall include the plural, the plural the singular, the use of any gender shall be applicable to all genders and the word "or" is used in the inclusive sense ("and/or"). When this Agreement refers to a number of days, unless otherwise specified as business days, that reference is to calendar days. The captions of this Agreement are for convenience of reference only and in no way define, describe, extend or limit the scope or intent of this Agreement or the intent of any provision contained in this Agreement. The term "including" as used herein shall mean including, without limiting the generality of any description preceding such term. A reference to a Schedule, Article, Section or clause is a reference to a Schedule, Article, Section or clause

of this Agreement, and the terms “hereof,” “herein,” and other like terms refer to this Agreement as a whole, including any attachments hereto. The language of this Agreement shall be deemed to be the language mutually chosen by the Parties and no rule of strict construction shall be applied in favor of or against either Party hereto.

11.9 Governing Law. This Agreement and the rights and obligations of the Parties shall be governed by and construed under the laws of the Commonwealth of Pennsylvania, excluding any conflicts or choice of law rule or principle that might otherwise refer construction or interpretation of this Agreement to the substantive law of another jurisdiction. Code of Federal Regulation parts and clauses shall be governed by Federal law, including the Federal common law of agreements.

11.10 Parties Independent. In making and performing this Agreement, the Parties act and shall act at all times as independent entities and nothing contained in this Agreement shall be construed or implied to create an agency, partnership or employer and employee relationship between NCDMM and Sub-Recipient. Except as expressly provided herein, at no time shall either Party make commitments or incur any charges or expenses for or in the name of the other Party.

11.11 Conflicts. Except as otherwise required by Applicable Law, to the extent that any provision of this Agreement conflicts with a provision of a Schedule, the provision of this Agreement shall govern.

11.12 Cumulative Remedies. Unless expressly set forth herein to the contrary, all remedies set forth herein are cumulative and are in addition to any and all remedies provided either Party at law or in equity.

11.13 Entire Agreement; Amendment. This Agreement and the previously signed America Makes (NAMII) Sub-Recipient/Membership agreement contain the entire understanding of the Parties with respect to the subject matter hereof. Any amendment or supplement to this Agreement shall be effective only if in writing signed by each Party.

11.14 Counterparts. This Agreement may be executed in any number of counterparts, each of which shall be an original as against the Party whose signature appears thereon, but all of which taken together shall constitute but one and the same instrument.

IN WITNESS WHEREOF, each Party has caused this Sub-Recipient Agreement to be signed by its duly authorized officer as of the date first above written.

| NCDMM                                                                                                                                   | [Sub-Recipient]                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| By: <b>Gene D Berkebile</b><br><small>Digitally signed by Gene D Berkebile<br/>Date: 2024.02.05 11:24:03 -05'00'</small><br>(Signature) | By: <br>(Signature) |
| Name: <b>Gene D Berkebile</b><br>(Printed Name)                                                                                         | Name: <b>Stephanie Gaffney</b><br>(Printed Name)                                                        |

Title: VP and CFO

Date: 2/5/2024

Title: VP of Advanced Manufacturing

Date: 22-2024

**SCHEDULE A**

**Youngstown Business Incubator (YBI) proposal titled Additive Manufacturing Training and all associated Exhibits are hereby incorporated by reference and in their entirety.**

**STATEMENT OF OBJECTIVES**

16 August 2023

|                                       |                                                |
|---------------------------------------|------------------------------------------------|
| <b>Project Title</b>                  | Additive Manufacturing Training                |
| <b>Funding Amount</b>                 | \$1,100,000                                    |
| <b>Funding Type</b>                   | FY23 CII, RDT&E                                |
| <b>Federal Funding Agency</b>         | OSD ManTech                                    |
| <b>Period of Performance (PoP)</b>    | 21 months from date of award + 2 for Reporting |
| <b>Cost Share (Objective/In Kind)</b> | TBD                                            |
| <b>ITAR Determination</b>             | Not ITAR                                       |

**Minimum Deliverables**

|                                                                 |                                                  |
|-----------------------------------------------------------------|--------------------------------------------------|
| Financial and Man-hour Report                                   | Monthly                                          |
| Technical Progress Report                                       | Quarterly                                        |
| Infrastructure Requirements Report                              | NLT 2 months after award                         |
| DRIVE AM Trainers                                               | At least 2 trainers by NLT 12 months after award |
| Customer discovery Report                                       | NLT 5 months after award                         |
| One SME portfolio                                               | NLT 4 months after award                         |
| One report on prior case studies                                | NLT 6 months after award                         |
| One report on manufacturing analysis methodology for case study | NLT 8 months after award                         |
| DRIVE AM course offerings in Northeast Ohio                     | NLT 22 months after award                        |
| Evaluation method for executive level trainings                 | NLT 12 months after award                        |
| 4 manufacturing method reports                                  | NLT 22 months after award                        |
| One digital tool for process selection                          | NLT 22 months after award                        |
| Executive level training courses                                | NLT 22 months after award                        |
| Delivery assessment report of the deployment to 5 SMM's         | NLT 22 months after award                        |
| Additional job roles on AMTrain                                 | NLT 22 months after award                        |
| Final Report                                                    | NLT 23 months after award                        |

**BACKGROUND**

This effort is focused on benefiting small and medium sized manufacturers (SMMs) in our Nation to significantly advance the manufacturing supply chain in the United States, as these SMMs typically do not have the financial or labor resources to implement advanced manufacturing strategies via trial-and-error methods. The effective selection of an appropriate manufacturing method requires training on the use of specific manufacturing tools, and perhaps more importantly, a deep understanding of the entire manufacturing method from design to inspection. DRIVE AM is a program focused on understanding these issues and has targeted training at developing the foundational knowledge for users and decision makers to understand the “realities” of additive manufacturing (AM), in particular, and how AM fits in the overall manufacturing supply chain, in general. This approach requires multi-day courses to cover the breadth and depth required. A major gap in this approach to date has been that it misses the executive level decision-makers, since decision-makers cannot commit to multiday trainings. Furthermore, most executives do not only need education in AM but would also benefit from a realistic and data-driven comprehensive education in overall manufacturing process selection to better enable

the defense supply chain to make the best decisions in selecting a manufacturing method for a particular application. DRIVE AM Youngstown will offer the established DRIVE AM courses but will also leverage the Ohio industry base to develop data-driven material/process selection training for executive/management level individuals. This will better allow SMM's to select the appropriate manufacturing methods for a particular application, and lead to SMM's pursuing improved education in specific manufacturing methods of interest beyond AM. Ultimately, this program will benefit northeastern Ohio SMM's and communities by educating best practices for development and integration of the appropriate manufacturing method (AM or otherwise) – leading to better decisions and decision-makers, a more capable workforce and an improved SMM supply chain in Northeast Ohio. The effort has identified four major tasks to include:

- Task 1. Deploy DRIVE AM course offerings in Northeast Ohio.
- Task 2. Develop technology adoption decision maker data for small to medium sized businesses in Northeast Ohio.
- Task 3. Develop data-driven decision maker/executive level educational resources.
- Task 4. Launch and evaluate DRIVE AM resources across Northeast Ohio.

The first major task is to expand the DRIVE AM course offerings developed previously and provide them in Northeast Ohio. Tasks 2-4 include:

- 1) gathering pedigree data on traditional and advanced manufacturing methods,
- 2) utilizing the data to create trainings and digital resources, and
- 3) performing trainings and evaluating their effectiveness.

The DRIVE AM team will take several realistic scenarios and test manufacturing process selection using constraints on time, cost, part design, materials, and more. The effort will leverage Ohio's manufacturing industry and the result will be a better-informed region/nation on the capabilities and constraints for AM and other manufacturing methods – benefiting from prior work focused on a single technology and applying best practices to a broad selection of technologies. This effort is intended to provide unbiased data that acknowledges all manufacturing methods have pros and cons and no single tool will be the right choice in all scenarios. As an example, one effort may include identification of a forged, high-strength aluminum alloy part for a military aircraft. The DRIVE AM team will utilize the defined approach to evaluate the design, manufacturing process, post forging finishing, and inspection methods. In addition, the team will purchase one of the parts and independent inspection to include cut ups for metallography.

Additional details for each task are provided below.

**Task 1. Deploy DRIVE AM course offerings in Northeast Ohio.**

- Evaluate potential training facility infrastructure against equipment needs to conduct training and/or identify other sufficient Northeast Ohio training partner facility (**Responsible: America Makes, YBI, UTEP**)
  - Acquire what is required to offer DRIVE AM foundational courses at identified Northeast Ohio partner, America Makes to facilitate leveraging Eastern Gateway Community College Additive Innovation Center on the Youngstown, OH campus
  - Evaluate and modify, as required, training content to better accommodate equipment availability at Eastern Gateway Community College Additive Innovation Center on the Youngstown, OH campus

- Deploy train-the-trainer program with YBI (or designated Northeast Ohio) personnel  
**(Responsible: America Makes, YBI, UTEP)**
  - Identify 2-5 individuals at the YBI campus (or with access to YBI campus) to become DRIVE AM trainers (Responsible: YBI)
    - Each individual will have to travel to the Keck Center for 3D Innovation at UTEP to become DRIVE AM trainers
- Offer DRIVE AM Foundation courses in Northeast Ohio at least 8 times **(Responsible: America Makes, YBI, UTEP, Tailored Alloys - First deployment will be co-delivered UTEP/YBI)**
  - Course offering include:
    - Introduction to Industrial Additive Manufacturing
    - Introduction to Additive Manufacturing Workflow
    - Laser Powder Bed Fusion Foundation (Tailored Alloys to deploy 2 times or as demand requires)
    - Material Extrusion Foundation
  - This effort will require marketing to fill seats and past efforts have shown this cost to be significant. Due to its proximity to America Makes home office, America Makes program managers, membership, and executive level visitors will be extended the opportunity to participate.

**Task 2. Develop technology adoption decision maker data for small to medium sized businesses in Northeast Ohio.**

- Evaluate manufacturing and process selection tools/guides/handbooks and deliver a report on findings **(Responsible: Tailored Alloys, YBI)**
- Perform customer discovery across a minimum of 20 SMMs from Northeast Ohio to determine gaps and preferences for education/training/data delivery. **(Responsible: YBI, Tailored Alloys, AMEP)**
  - Leverage existing and past America Makes efforts to supplement comparisons and case studies
  - Deliver one report on customer discovery of at least 20 Northeast Ohio manufacturing companies
- Develop a methodology for unbiased and comprehensive evaluation of metal part manufacturing selection to include design, material, post processing, inspection, criticality, workforce, and sustainment **(Responsible: Tailored Alloys, AMEP, YBI)**
  - Create DRIVE AM manufacturing team to include forging, casting, machining, additive manufacturing subject matter experts
  - Deliver one DRIVE AM subject matter expert team portfolio
  - Deliver one report detailing manufacturing analysis methodology for case study
- Perform a series of manufacturing comparison tests to develop foundational knowledge on costs, quality, and time to include LPBF, Casting, Machining, and Forging at a minimum. **(Responsible: YBI, UTEP, Tailored Alloys)**
  - Identify candidate parts from DoD supply chain and northeast Ohio businesses
    - This may include the purchase of rights to technical data packages from suppliers to facilitate participation in the program
    - Evaluate parts by acquiring pedigree data to be used for data-driven decision-making approach
    - At a minimum, one part will be taken from design to deployment

- Analysis of parts to be performed by UTEP (metallography and mechanical, and XCT if it permits)
- Produce 4 completed manufacturing method reports to include a Casting, Forging, Machining, and metal AM part **(Responsible: YBI, Tailored Alloys, AMEP)**

**Task 3. Develop data-driven decision maker/executive level educational resources.**

- Utilize data developed in Task 1 to create decision maker educational resources **(Responsible: UTEP, Tailored Alloys)**
  - Create short executive trainings that are based on direct feedback from SMMs, test response to content, perform CQI, and continue delivery
    - Develop 4 different executive level in person trainings from 30min to 8hours
    - Develop accompanying evaluation methods to determine effectiveness
    - Utilize data developed in Task 1 and feedback from SMMs during course development
    - Work with UTEP to test response to content and perform CQI
- Perform the executive level in person trainings developed by DRIVE AM team to at least 40 individuals **(Responsible: America Makes, YBI, Tailored Alloys)**
  - Perform trainings of at least 20 people per training method to conduct analysis. **(Responsible: YBI, Tailored Alloys)**
  - Work with UTEP to test response to content and perform CQI

**Task 4. Launch DRIVE AM resources across Northeast Ohio.**

- Down select up to five businesses across Northeast Ohio to participate in DRIVE AM program **(Responsible: YBI, DRIVE AM 501c3, UTEP, Tailored Alloys)**
  - Assess individual knowledge gaps across each business
  - Demonstrate how DRIVE AM solutions can be easily pulled from resources and deployed by SMM's
  - Introduce DRIVE AM resources to improve operations
- Perform training and onsite detailed analysis of 5 manufacturing facilities in Northeast Ohio to determine if and how advanced manufacturing could be incorporated to support DoD supply chain. **(Responsible: America Makes, YBI, UTEP, Tailored Alloys)**
  - Perform evaluation of content for continual improvement and ensure effectiveness

The specific deliverables for the effort include:

- Complete infrastructure enhancement at YBI or to offer DRIVE AM foundational courses **(Responsible: YBI, UTEP)**
- Train at least two YBI (or designated) staff to be DRIVE AM trainers **(Responsible: YBI, UTEP)**
- Hold at least 8 DRIVE AM course offerings in Northeast Ohio (e.g., at YBI) **(Responsible, YBI, UTEP, Tailored Alloys - First deployment will be co-delivered UTEP/YBI)**
- One report detailing manufacturing analysis methodology for case study **(Responsible: Tailored Alloys, AMEP, YBI)**
- One DRIVE AM subject matter expert team portfolio **(Responsible: Tailored Alloys, AMEP, YBI)**
- 4 completed manufacturing method reports to include a Casting, Forging, Machining, and metal AM part **(Responsible: YBI, Tailored Alloys, AMEP)**

- One report on prior case studies comparing manufacturing methods **(Responsible: YBI, AMEP, Tailored Alloys)**
- One report on customer discovery of at least 20 Northeast Ohio manufacturing companies **(Responsible: YBI, Tailored Alloys, AMEP)**
- Develop 4 different executive level in person trainings from 30min to 8hours and accompanying evaluation methods to determine effectiveness **(Responsible: Tailored Alloys, UTEP)**
  - Perform trainings of at least 20 people per training method to conduct analysis. **(Responsible: YBI, Tailored Alloys)**
- Perform training and onsite detailed analysis of 5 manufacturing facilities in Northeast Ohio to determine if and how advanced manufacturing could be incorporated to support DoD supply chain. **(Responsible: YBI, UTEP, Tailored Alloys)**

**Key Performance Parameters**

| ITEM | KPP                                                                | DESCRIPTION                                                                                                            | THRESHOLD | TARGET VALUE | UNIT OF MEASURE             |
|------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------|--------------|-----------------------------|
| 1    | Resources evaluated for Process selection                          | Number of resources evaluated for manufacturing process selection                                                      | 15        | 30           | Recourses Evaluated         |
| 2    | DRIVE AM trainers in northeast Ohio                                | Persons in northeast Ohio having completed UTEP's Train-the-Trainer Program                                            | 2         | 5            | persons                     |
| 3    | DRIVE AM trainings in northeast Ohio                               | DRIVE AM trainings delivered by UTEP (or its trained trainers) in northeast Ohio                                       | 6         | 7            | training courses            |
| 4    | Evaluated parts                                                    | Evaluate parts to support case studies that compare forging, machining, casting, and LPBF                              | 3         | 4            | parts                       |
| 5    | Executive level trainings                                          | Develop trainings targeting executives making decisions on manufacturing use                                           | 2         | 4            | trainings                   |
| 6    | Evaluation of tailored knowledge base across 5 SMMs                | DRIVE AM, Inc will launch and evaluate the effectiveness of tailored digital resources                                 | 5         | 10           | Per SMM                     |
| 7    | Create user guides to complement technical training solutions      | DRIVE AM, Inc. will create user guides to ensure industry can more easily adopt technical training solutions developed | 3         | 5            | Per user guide created      |
| 8    | Deploy technical training solutions to 3 SMMs to demonstrate value | DRIVE AM, Inc will deploy one or all of the solutions to machines currently being used in industry                     | 3         | 10           | Per solution deployed       |
| 9    | Develop a tailored knowledge base for 5 SMMs                       | DRIVE AM, Inc. will develop a tailored knowledge base for SMM in Ohio                                                  | 5         | 10           | Per tailored knowledge base |
| 10   | Additional job roles on AMTrain                                    | Additional job roles/competencies mapped                                                                               | 20        | 24           | Number of job roles         |

## Minimum Deliverables

|                                                                 |                                                  |
|-----------------------------------------------------------------|--------------------------------------------------|
| Financial and Man-hour Report                                   | Monthly                                          |
| Technical Progress Report                                       | Quarterly                                        |
| Infrastructure Requirements Report                              | NLT 2 months after award                         |
| DRIVE AM Trainers                                               | At least 2 trainers by NLT 12 months after award |
| Customer discovery Report                                       | NLT 5 months after award                         |
| One SME portfolio                                               | NLT 4 months after award                         |
| One report on prior case studies                                | NLT 6 months after award                         |
| One report on manufacturing analysis methodology for case study | NLT 8 months after award                         |
| DRIVE AM course offerings in Northeast Ohio                     | NLT 22 months after award                        |
| Evaluation method for executive level trainings                 | NLT 12 months after award                        |
| 4 manufacturing method reports                                  | NLT 22 months after award                        |
| One digital tool for process selection                          | NLT 22 months after award                        |
| Executive level training courses                                | NLT 22 months after award                        |
| Delivery assessment report of the deployment to 5 SMM's         | NLT 22 months after award                        |
| Additional job roles on AMTrain                                 | NLT 22 months after award                        |
| Final Report                                                    | NLT 23 months after award                        |

## Project Timeline

| <div><div><div></div><div>YBI</div></div><div>YOUNGSTOWN<br/>BUSINESS INCUBATOR</div></div> |                                                                     | Task                                                                                   | Month |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--|
|                                                                                             |                                                                     |                                                                                        | 1     | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |  |
|                                                                                             |                                                                     | PROJECT KICKOFF                                                                        |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| 1 - Deploy                                                                                  |                                                                     | DRIVE AM Course Offerings in NE Ohio                                                   |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|                                                                                             | Evaluate YBI Infrastructure against equipment needs                 |                                                                                        |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|                                                                                             | Train the trainer program                                           |                                                                                        |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|                                                                                             | DRIVE AM courses in NE Ohio (8 courses)                             |                                                                                        |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| 2 - Develop                                                                                 |                                                                     | technology adoption decision maker data for mall to medium sized businesses in NE Ohio |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|                                                                                             | Evaluate manufacturing and process selection tools/guides/handbooks |                                                                                        |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|                                                                                             | Customer discovery to determine gaps and preferences                |                                                                                        |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|                                                                                             | Methodology development for evaluation of metal part manufacturing  |                                                                                        |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|                                                                                             | Manufacturing comparison tests                                      |                                                                                        |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|                                                                                             | Complete manufacturing method reports                               |                                                                                        |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| 3 - Develop                                                                                 |                                                                     | data-driven decision maker/executive level educational resources.                      |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|                                                                                             | Create decision maker educational resources from Task 1 data        |                                                                                        |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|                                                                                             | In-person training                                                  |                                                                                        |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| 4 - Launch                                                                                  |                                                                     | DRIVE AM resources across NE Ohio                                                      |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|                                                                                             | Business downselection for DRIVE AM participation                   |                                                                                        |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|                                                                                             | Training and onsite analysis of 5 manufacturing facilities          |                                                                                        |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|                                                                                             |                                                                     | FINAL REPORTING                                                                        |       |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |

**OPERATIONS SECURITY (OPSEC)**

The recipient shall participate in all activities associated with the disciplines of the organization's Industrial Security, Information Security, Personnel Security, Operations Security (OPSEC), and Antiterrorism programs, following appropriate measures in each program as required for this particular agreement. Security measures are required to reduce program vulnerability from successful adversary collection, exploitation of critical information, and violations of export control requirements. The prime recipient shall ensure all subrecipients, if applicable, conform to these requirements as required by the prime recipient.

**PROGRAM PROTECTION PLAN (PPP) REQUIREMENTS**

Any potential critical program information (CPI) generated as part of this effort will be reviewed to determine the need for a PPP or to be included as part of an existing PPP.

**SCHEDULE B**

**BUDGET**

**Team Member Name: Youngstown Business Incubator**

**Proposal Number: DriveAM with UTEP and Tailored Alloys 24 month budget**

|                                               | <b>GRAND<br/>TOTAL</b> |
|-----------------------------------------------|------------------------|
| REQUESTED AMERICA MAKES FEDERAL FUNDING       |                        |
| LABOR                                         | 583,594                |
| TRAVEL                                        | 40000                  |
| SUBCONTRACT                                   | 0                      |
| MATERIALS                                     | 5000                   |
| EQUIPMENT                                     | 0                      |
| CONSULTANT                                    | 60000                  |
| ODC's                                         | 415000                 |
| TOTAL REQUESTED AMERICA MAKES FEDERAL FUNDING | 1,103,594              |
| COST SHARE                                    |                        |
| LABOR                                         | 0                      |
| TRAVEL                                        | 0                      |
| SUBCONTRACT                                   | 0                      |
| MATERIALS                                     | 0                      |
| EQUIPMENT                                     | 0                      |
| CONSULTANT                                    | 0                      |
| ODC's                                         | 0                      |
| TOTAL PROPOSED COST SHARE                     | 0                      |
| TOTAL PROJECT EFFORT                          | 1,103,594              |

## SCHEDULE C

### APPLICABLE FEDERAL REGULATION AND PRIME AGREEMENT FLOW-DOWN

#### **1.00 Administrative Requirements and Order of Precedence (Mar 2015)**

(a) This award is governed by the guidance in 2 Code of Federal Regulations (CFR) part 200, "Uniform Administrative Requirements, Cost Principles, and Audit Requirements for Federal Awards" as modified and supplemented by the Department of Defense's (DoD) interim implementation found at 2 CFR part 1103, "Interim Grants and Cooperative Agreements Implementation of guidance in 2 CFR part 200" (79 FR 76047, December 19, 2014), all of which are incorporated herein by reference. Provisions of Chapter I, Subchapter C of Title 32, CFR, "DoD Grant and Agreement Regulations," other than part 33, continue to be in effect and are incorporated herein by reference, with applicability as stated in those provisions.

(b) In the event of a conflict between the terms of this agreement and other government documents the conflict shall be resolved by giving precedence in descending order as follows:

- (1) Federal statutes
- (2) Federal regulations
- (3) 2 CFR part 200, as modified and supplemented by DoD's Interim implementation found in 2 CFR part 1103
- (4) Award-specific terms and conditions
- (5) Attachments to this award, if any
- (6) In case of disagreement with any requirements of this award, the recipient shall contact the NCDMM in order to resolve the issue. The recipient shall not access any costs to the award or accept payments until the issue is resolved.

#### **1.01 Trafficking in Persons (March 2015)**

This award is subject to the requirements of section 106 (g) of the Trafficking Victims Protection Act of 2000 (22 U.S.C. 7104, as implemented by 2 CFR 175)

You as the sub-recipient and your employees may not—

- (a) Engage in severe forms of trafficking in persons during the period of time that the award is in effect; or
  - (ii) Procure a commercial sex act during the period of time that the award is in effect; or
  - (iii) Use forced labor in the performance of the award or subawards under the award.

#### **1.02 Title to Property – Identified in Budget (March 2015)**

Title to personal property acquired with agreement funds shall vest in the sub-recipient upon acquisition, except that supplies shall be managed in accordance with 2 CFR 200 Section 314. Title to real property shall vest in the sub-recipient subject to conditions contained in 2 CFR 200 Section 311. The sub-recipient shall dispose of real property in accordance with 2 CFR 200 Section 311.

#### **1.03 Property System (March 2015)**

Sub-recipient's property system shall meet the standards as set forth in 2 CFR 200 Section 310-316.

#### **1.04 Cost Principles (March 2015)**

Sub-recipient(s) shall comply with the cost principles as contained in 2 CFR 200, Subpart E, Cost Principles. Cost principles for determining allowability of costs applicable to lower tier, cost type contracts or awards under this agreement shall be determined by type of entity receiving the award.

### **1.05 Standards for Financial Management (March 2015)**

Sub-recipient(s) financial management system shall comply with the standards identified in 2 CFR 200 Section 302.

### **1.06 Audit Requirements/Retention and Access to Records (March 2015)**

Sub-recipient shall comply with the audit requirements of 2 CFR 200, Subpart F Audit Requirements and shall comply with the requirements appropriate for the type of entity receiving the award. Sub-recipient's financial records, supporting documents, statistical records, and all other records pertinent to an award shall be retained and access to them permitted in accordance with 2 CFR 200 Section 336.

### **1.07 Cost Sharing (August 2001)**

Sub-recipient's contributions may count as cost sharing only to the extent that they comply with 2 CFR 200(200.306)

### **1.08 Export Control (March 2015) (TAILORED)**

(a) Access to the technology developed under this agreement by foreign firms, institutions or individuals shall be controlled by the Sub-recipient under applicable U.S. export control laws.

(b) The sub-recipient shall receive NCDMM and grants officer approval before assigning or granting access to any work, equipment, or technical data generated or delivered under this agreement to foreign persons or their representatives. The notification shall include the name and country of origin of the foreign person or representative, the specific work, equipment, or data to which the person will have access.

### **1.09 Inventions (March 2015)**

(a) The clause entitled Standard Patent Rights, (37 CFR 401.14(a)) is hereby incorporated by reference and is modified as follows: replace the word "recipient" with "Sub-recipient"; replace the words "agency," "Federal Agency" and "funding Federal Agency" with "NCDMM"; replace the word "contract" with "agreement"; delete paragraphs (g)(2), (g)(3) and paragraph (g)(1) first sentence to read "The recipient will include this clause for experimental, developmental or research work to be performed by a Sub-recipient." Paragraph (1), Communications, point of contact on matters relating to this clause will be the servicing Staff Judge Advocate's office.

(b) Interim or final Invention Reports 1) listing subject invention(s) and stating that all subject inventions have been disclosed, or 2) stating that there are no such inventions, shall be sent to the NCDMM. Please include in the subject line of the e-mail the contract number followed by the words "Invention Reporting." The Sub-recipient shall file Invention (Patent) Reports on the DD Form 882, Report of Inventions and Subcontracts, as of the close of each performance year and at the end of the term for this agreement. Annual reports are due 60 days after the end of each year of performance and final reports are due 60 days after the expiration of the final performance period. Negative reports are also required annually.

(c) The DD Form 882 may also be used for the notification any lower tier award for experimental, developmental or research work which contain a "Patent Rights" clause

(d) This provision also constitutes the request for the following information for any subject invention for which the recipient has retained ownership: 1) the filing date, 2) serial number and title, 3) a copy of the patent application and 4) patent number and issue date. Submittal shall be to the NCDMM.

### **1.10 Data Rights (March 2015)**

(a) All rights and title to data, as defined in 48 CFR 27.401, generated under this agreement shall vest in the sub-recipient.

(b) The sub-recipient hereby grants to the U.S. Government a royalty free, world-wide, nonexclusive, irrevocable license to use, modify, reproduce, release, perform, display or disclose any data for Government purposes.

- "Government purpose includes competitive procurement, but not the rights to use, modify, reproduce, release, perform, display, or disclose technical data for commercial purposes or authorize others to do so"

(c) The sub-recipient is responsible for affixing appropriate markings indicating rights on all data delivered under the agreement. The Government will have unlimited rights in all data delivered without markings.

(d) The sub-recipient shall include this article, suitably modified to identify the parties, in all lower tier contracts and awards, regardless of tier, for experimental, developmental, or research work.

### **1.11 Publishing Project Results (March 2015)**

(a) Publications. The Sub-recipient(s) is expected to publish or otherwise make publicly available the results of the work conducted under this subaward. One copy of all publications resulting from the project shall be forwarded to the NCDMM as it becomes available.

(b) An acknowledgment of awarding agency's support shall appear in the publication of any material, whether copyrighted or not. The acknowledgement shall read:

"This material is based on research sponsored by Air Force Research Laboratory under agreement number FA8650-20-2-5700. The U.S. Government is authorized to reproduce and distribute reprints for Governmental purposes notwithstanding any copyright notation thereon."

(c) The Sub-recipient is responsible for assuring that every publication of material based on or developed under this project contains the following disclaimer:

"The views and conclusions contained herein are those of the authors and should not be interpreted as necessarily representing the official policies or endorsements, either expressed or implied, of Air Force Research Laboratory or the U.S. Government."

### **1.12 Reporting Subawards and Executive Compensation (March 2015)**

(a) Reporting of first-tier subawards.

(1) Applicability. Unless you are exempt as provided in paragraph d. of this award term, you must report each action that obligates \$25,000 or more in Federal funds that does not include Recovery funds (as defined in section 1512(a)(2) of the American Recovery and Reinvestment Act of 2009, Pub. L. 111-5) for a subaward to an entity (see definitions in paragraph e. of this award term).

(2) Where and when to report.

(i) You must report each obligating action described in paragraph a.1. of this award term to <http://www.fsrs.gov>.

(ii) For subaward information, report no later than the end of the month following the month in which the obligation was made. (For example, if the obligation was made on November 7, 2010, the obligation must be reported by no later than December 31, 2010.)

(3) What to report. You must report the information about each obligating action that the submission instructions posted at <http://www.fsr.gov> specify.

(b) Reporting Total Compensation of Recipient Executives.

(1) Applicability and what to report. You must report total compensation for each of your five most highly compensated executives for the preceding completed fiscal year, if-

(i) the total Federal funding authorized to date under this award is \$25,000 or more; (ii)

in the preceding fiscal year, you received-

(A) 80 percent or more of your annual gross revenues from Federal procurement

contracts (and subcontracts) and Federal financial assistance subject to the

Transparency Act, as defined at 2 CFR

170.320 (and subawards); and

(B) \$25,000,000 or more in annual gross revenues from Federal procurement contracts (and subcontracts) and Federal financial assistance subject to the Transparency Act, as defined at 2 CFR 170.320 (and subawards); and

(iii) The public does not have access to information about the compensation of the executives through periodic reports filed under section 13(a) or 15(d) of the Securities Exchange Act of 1934 (15 U.S.C. 78m(a), 78o(d)) or section 6104 of the Internal Revenue Code of 1986. (To determine if the public has access to the compensation information, see the U.S. Security and Exchange Commission total compensation filings at <http://www.sec.gov/answers/execomp.htm>.)

(2) Where and when to report. You must report executive total compensation described in paragraph b.1. of this award term:

(i) As part of your registration profile at <http://www.ccr.gov>.

(ii) By the end of the month following the month in which this award is made, and annually thereafter.

(c) Reporting of Total Compensation of Subrecipient Executives.

(1) Applicability and what to report. Unless you are exempt as provided in paragraph d. of this award term, for each first-tier subrecipient under this award, you shall report the names and total compensation of each of the subrecipient's five most highly compensated executives for the subrecipient's preceding completed fiscal year, if-

(i) in the subrecipient's preceding fiscal year, the subrecipient received

(A) 80 percent or more of its annual gross revenues from Federal procurement contracts (and subcontracts) and Federal financial assistance subject to the Transparency Act, as defined at 2 CFR 170.320 (and subawards); and

(B) \$25,000,000 or more in annual gross revenues from Federal procurement contracts (and subcontracts), and Federal financial assistance subject to the Transparency Act (and subawards); and

(ii) The public does not have access to information about the compensation of the executives through periodic reports filed under section 13(a) or 15(d) of the Securities Exchange Act of 1934 (15 U.S.C. 78m(a), 78o(d)) or section 6104 of the Internal Revenue Code of 1986. (To determine if the public has access to the compensation information, see the U.S. Security and Exchange Commission total compensation filings at <http://www.sec.gov/answers/excomp.htm>.)

(2) Where and when to report. You must report subrecipient executive total compensation described in paragraph c.1. of this award term:

(i) To the recipient.

(ii) By the end of the month following the month during which you make the subaward. For example, if a subaward is obligated on any date during the month of October of a given year (i.e., between October 1 and 31), you must report any required compensation information of the subrecipient by November 30 of that year.

(d) Exemptions: If, in the previous tax year, you had gross income, from all sources, under \$300,000, you are exempt from the requirements to report:

(1) Subawards, and

(2) The total compensation of the five most highly compensated executives of any subrecipient. (e)

Definitions. For purposes of this award term:

(1) Entity means all of the following, as defined in 2 CFR part 25:

(i) A Governmental organization, which is a State, local government, or Indian tribe;

(ii) A foreign public entity;

(iii) A domestic or foreign nonprofit organization; (iv)

A domestic or foreign for-profit organization;

(v) A Federal agency, but only as a subrecipient under an award or subaward to a non-Federal entity. (2)

Executive means officers, managing partners, or any other employees in management positions. (3)

Subaward:

(i) This term means a legal instrument to provide support for the performance of any portion of the substantive project or program for which you received this award and that you as the recipient award to an eligible subrecipient.

(ii) A subaward may be provided through any legal agreement, including an agreement that you or a subrecipient considers a contract.

(4) Subrecipient means an entity that:

- (i) Receives a subaward from you (the recipient) under this award; and
- (ii) Is accountable to you for the use of the Federal funds provided by the subaward.

(5) Total compensation means the cash and noncash dollar value earned by the executive during the recipient's or subrecipient's preceding fiscal year and includes the following (for more information see 17 CFR 229.402(c)(2)):

- (i) Salary and bonus.
- (ii) Awards of stock, stock options, and stock appreciation rights. Use the dollar amount recognized for financial statement reporting purposes with respect to the fiscal year in accordance with the Statement of Financial Accounting Standards No. 123 (Revised 2004) (FAS 123R), Shared Based Payments.
- (iii) Earnings for services under non-equity incentive plans. This does not include group life, health, hospitalization or medical reimbursement plans that do not discriminate in favor of executives, and are available generally to all salaried employees.
- (iv) Change in pension value. This is the change in present value of defined benefit and actuarial pension plans.
- (v) Above-market earnings on deferred compensation which is not tax-qualified.
- (vi) Other compensation, if the aggregate value of all such other compensation (e.g. severance, termination payments, value of life insurance paid on behalf of the employee, perquisites or property) for the executive exceeds \$10,000.

### **1.13 Disclosure or Information (March 2015) (Tailored)**

(a) The Sub-recipient shall not release to anyone outside the Sub-recipient's organization any unclassified information, regardless of medium (e.g., film, tape, document, media announcements, etc.), pertaining to U.S. Government Agency-Driven Projects unless-

- (1) The NCDMM Agreements Officer has given prior written approval; or
- (2) The information is otherwise in the public domain before the date of release.

(b) Requests for approval shall identify the specific information to be released, the medium to be used, and the purpose for the release. The Sub-recipient shall submit its request to the NCDMM Agreements Officer at least 65 days before the proposed date for release through NCDMM.

(c) The Sub-recipient agrees to include a similar requirement in each sub-agreement under this agreement. Sub-recipients shall submit requests for authorization to release through the NCDMM to the Agreements Officer.

### **1.14 Procurement System (March 2015)**

The sub-recipient's procurement system shall comply with the standards contained in 2 CFR 200 Section 317-326.

### **1.15 Military Recruiting on Campus (March 2015)**

As a condition for receipt of funds available to the Department of Defense (DoD) under this award, the Sub-recipient agrees that it is not an institution of higher education (as defined in 32 CFR 216) that has a policy or practice that prohibits or, in effect, prevents ROTC or military recruiting on campus, as described in DFARS 252.209-7005(b). If the Sub-recipient is determined, using the procedures in 32 CFR part 216, to be such an institution of higher education during the period of performance of this agreement, and therefore to be in breach of this clause, the NCDMM will cease all payments of DoD funds under this agreement and all other DoD grants and cooperative agreements to the sub-recipient, and it may suspend or terminate such grants and agreements unilaterally for material failure to comply with the terms and conditions of award.

### **1.16 Assurances (March 2015)**

(a) By signing or accepting funds under the agreement, the Sub-recipient assures that it will comply with applicable provisions of the following National policies on: (1) Prohibiting discrimination:

(i) On the basis of race, color, or national origin, in Title VI of the Civil Rights Act of 1964 (42 U.S.C. 2000d, et seq.), as implemented by DoD regulations at 32 CFR part 195;

(ii) On the basis of age, in the Age Discrimination Act of 1975 (42 U.S.C. 6101, et seq.) as implemented by Department of Health and Human Services regulations at 45 CFR part 90;

(iii) On the basis of handicap, in Section 504 of the Rehabilitation Act of 1973 (29 U.S.C. 794), as implemented by Department of Justice regulations at 28 CFR part 41 and DoD regulations at 32 CFR part 56;

(iv) On the basis of sex or blindness, in Title IX of the Educational Amendments of 1972 (20 U.S.C. 1681, et seq.).

(2) The Clean Air Act (42 U.S.C. 7401, et seq.) and Clean Water Act (33 U.S.C. 1251, et seq.), as implemented by Executive Order 11738 (3 CFR, 1971-1975 Comp., p. 799).

The Sub-recipient shall obtain assurances of compliance for all subawards under this effort

### **1.17 Prohibition on Using Funds under Grants and Cooperative Agreement with Entities that Require Certain Internal Confidentiality Agreements (January 2017)**

- (a) The Sub-recipient shall not require its employees or subawardees to sign or comply with internal confidentiality agreements or statements prohibiting or otherwise restricting such employees or subawardees from lawfully reporting waste, fraud, or abuse related to the performance of a Government assistance instrument to a designated investigative or law enforcement representative of a Federal department or agency authorized to receive such information (e.g., agency Office of the Inspector General).
- (b) The Sub-Recipient shall notify current employees and subawardees that prohibitions and restrictions of any preexisting internal confidentiality agreements or statements covered by this article, to the extent that such prohibitions and restrictions are inconsistent with the prohibitions of this article, are no longer in effect.
- (c) The prohibition in paragraph (a) of this clause does not contravene requirements applicable to Standard Form 312 (Classified Information Nondisclosure Agreement), Form 4414 (Sensitive Compartmented Information Nondisclosure Agreement), or any other form issued by a Federal department or agency governing the nondisclosure of classified information.
- (d) In accordance with section 743 of Division E, Title VII, of the Consolidated and Further Continuing Appropriations Act, 2015, (Pub. L. 113-235), and its successor provisions in subsequent appropriations acts (and as extended in continuing resolutions) use of funds appropriated (or otherwise made available) is prohibited, if the Government determines that the Sub-recipient is not in compliance with the provisions of this article.

(e) The Sub-recipient shall include the substance of this clause, including this paragraph (e), in subawards.

#### **1.18 Section 889 (August 2020)**

Section 889 of the National Defense Authorization Act (NDAA) for Fiscal Year (FY) 2019 (Public Law 115-232) prohibits the head of an executive agency from obligating or expending loan or grant funds to procure or obtain, extend, or renew a contract to procure or obtain, or enter into a contract (or extend or renew a contract) to procure or obtain the equipment, services, or systems prohibited systems as identified in section 889 of the NDAA for FY 2019.

To implement this requirement, in accordance with 2 CFR 200.216 and 200.471, DoD Grants Officers shall include a condition in all awards that are issued on or after August 13, 2020.

The condition shall include language enacting section 889 of the NDAA for FY 2019 by prohibiting the recipient and subrecipient from entering into contracts (or extending or renewing contracts) with entities that use covered telecommunications equipment or services. This prohibition shall apply even if the contract is not intended to procure or obtain any equipment, system, or service that uses covered telecommunications equipment or services.

For the purposes of this condition,

1. COVERED TELECOMMUNICATIONS EQUIPMENT OR SERVICES means any of the following:

- \* For the purpose of public safety, security of government facilities, physical security surveillance of critical infrastructure, and other national security purposes, video surveillance and telecommunications equipment produced by Hytera Communications Corporation, Hangzhou Hikvision Digital Technology Company, or Dahua Technology Company (or any subsidiary or affiliate of such entities);
- \* Telecommunications or video surveillance equipment or services provided by such entities or using such equipment; or
- \* Telecommunications or video surveillance equipment or services produced or provided by an entity that the Secretary of Defense, in consultation with the Director of the National Intelligence or the Director of the Federal Bureau of Investigation, reasonably believes to be an entity owned or controlled by, or otherwise connected to, the government of a foreign country.

2. COVERED FOREIGN COUNTRY means the People's Republic of China.

#### **1.19 Prohibition on a ByteDance Covered Application (June 2023)**

(a) *Definitions.* As used in this article—

*Covered application* means the social networking service TikTok or any successor application or service developed or provided by ByteDance Limited or an entity owned by ByteDance Limited.

*Information technology*, as defined in 40 U.S.C. 11101(6)—

(1) Means any equipment or interconnected system or subsystem of equipment, used in the automatic acquisition, storage, analysis, evaluation, manipulation, management, movement, control, display, switching, interchange, transmission, or reception of data or information by the executive agency, if the equipment is used by the executive agency directly or is used by an awardee under an

award with the executive agency that requires the use—

(i) Of that equipment; or

(ii) Of that equipment to a significant extent in the performance of a service or the furnishing of a product;

(2) Includes computers, ancillary equipment (including imaging peripherals, input, output, and storage devices necessary for security and surveillance), peripheral equipment designed to be controlled by the central processing unit of a computer, software, firmware and similar procedures, services (including support services), and related resources; but

(3) Does not include any equipment acquired by a Federal awardee incidental to a Federal award.

- (b) *Prohibition.* Section 102 of Division R of the Consolidated Appropriations Act, 2023 (Pub. L. 117-328), the No TikTok on Government Devices Act, and its implementing guidance under Office of Management and Budget (OMB) Memorandum M-23-13, dated February 27, 2023, “No TikTok on Government Devices” Implementation Guidance, collectively prohibit the presence or use of a covered application on executive agency information technology, including certain equipment used by Federal awardee(s). The Awardee is prohibited from having or using a covered application on any information technology owned or managed by the Government, or on any information technology used or provided by the Awardee under this award, including equipment provided by the Awardee’s employees; however, this prohibition does not apply if the Grants Office/Agreements Officer provides written notification to the Awardee that an exception has been granted in accordance with OMB Memorandum M-23-13.
- (c) *Subcontracts/Subrecipients.* The Awardee shall insert the substance of this article, including this paragraph (c), in all subawards, including subawards for the acquisition of commercial products or commercial services.

## **Schedule D**

### **Additional Requirements**

#### **Kickoff Meeting**

The sub-recipient will schedule a kickoff meeting with the NCDMM Project Manager to be held within 30 days of contract award. At the kickoff meeting the sub-recipient will provide the following:

- A detailed Gantt chart for the entire project that includes clearly identified milestones. The milestones must include identification of the key elements of Technology Development, Education and Workforce, Technology Transition, Sustainability, and MRL/TRL status.
- A listing of project deliverables with associated delivery dates
- A Spend Plan by month for the entire project. The attached template is provided as assistance. The Spend Plan must incorporate both America Makes funding and cost share.
- Identification of team Sub-Recipient/Members and their roles and responsibilities

#### **Project Reporting**

For the duration of the project, monthly reports, invoices and cost share reports will be due by the 10<sup>th</sup> of the month for activities of the preceding month. The sub-recipient is responsible for assembling information from subcontractors to support reporting requirements. Both technical and financial reports will be required.

- Technical reports will highlight the reporting period's technical achievements, identify any unexpected technical issues that have been encountered and action plans to resolve such issues, state the current MRL/TRL level and justification for advancing in MRL/TRL, updated Gantt charts, progress on metrics and future planned activities. All technical reports must provide a brief publicly-releasable section and a section for release to America Makes Sub-Recipient/Members. Proprietary information available only to project team Sub-Recipient/Members and appropriate America Makes staff/Government representatives can also be contained in the report and marked accordingly. A template for the reporting format will be provided by NCDMM/America Makes.
- Financial reports and invoices will document expenditures of both America Makes funds and cost share. Pertinent backup documentation, such as cost share reports, will be required to support financial reports. Backup documentation to be supplied by the sub-recipient will include, but not be limited to, individual cost share reports for team Sub-Recipient/Members. Tracking of actual expenditures vs. those provided in the Spend Plan will be required. The ratio of America Makes funding vs. cost share funding is expected to be roughly 1:1 at all times throughout the project. Both the ratio of America Makes funding to cost share funding and actual expenditures to expenditures stated in the Spend Plan will be monitored by the NCDMM Project Manager. If either or both of these financial goals are not met, the sub-recipient will be required to submit a corrective action plan. Failure to correct financial shortcomings may be grounds for contract cancellation.
- America Makes will provide an online portal for reporting. All reports will be uploaded to the America Makes online portal site by the sub-recipient. Note: Microsoft SharePoint will be used initially and log-on credentials will be provided by the America Makes team.

#### **Meeting Participation**

- Sub-recipient participation in regularly scheduled conference calls, quarterly review meetings and Project Management Review (PMR) meetings will be required.